Chapter 6: Quantitative Data Analysis Techniques

Univariate and Bivariate Analysis, Measures of Central Tendency (Section 1)

What do quantitative researchers analyse? Types of statistics

Descriptive statistics: Computations that describe characteristics or relationships among variables (Limited to the data set and does not look to draw conclusions about groups outside the data set. Main tools/metrics include measures of central tendency, standard deviations; depicted through distribution tables, histograms, pie charts, etc.)

Inferential statistics: Computations for making inferences about samples for the larger populations (Uses data in the set to make inferences/predictions for larger populations. Main tools/metrics include hypothesis tests, confidence intervals, etc.)

Understanding univariate vs bivariate analysis in quantitative research

While univariate analysis explains the individuals in a group of study (units of analysis), bivariate analysis emphasizes the explanation of relationships between variables

Bivariate analysis emphasizes the sub-group comparisons within a sample itself

Bivariate analysis helps the researcher understand the association between variables, with the variables being of the same sample

Univariate analysis

- Used to describe single variables only

A simple example of univariate analysis is to examine the salaries of employees in a company to find the average, median, and range of salaries. This analysis focuses on a single variable, in this case, salary, to understand its distribution within the group using measures like central tendency (mean, median) and dispersion (range, variance)

Measures of central tendency

- Measures of central tendency are single-value summary statistics that describe the typical or central value of a dataset, with the most common ones being the mean, median, and mode.
- The mean is the sum of all values divided by the count of values (average)
- The median is the middle value in an ordered dataset (mid value)
- The mode is the most frequently occurring value in a dataset (most repeated)

Measures of central tendency

Mean

- Definition: The arithmetic average of a dataset.
- How to calculate: Sum all the values in the dataset and divide by the total number of values.
- Example: For the data {2, 3, 4, 5}, the mean is (2+3+4+5)/4 = 3.5.

Important to remember;

The mean can not be used for categorical data (non-numerical data like colours, names, etc.) but it is impacted by outliers in a data set.

Measures of central tendency

Median

- Definition: The middle value when the dataset is arranged in order from smallest to largest.
- How to calculate:
  - Order the data.
  - If there is an odd number of data points, the median is the single middle value.
  - If there is an even number of data points, the median is the average of the two middle values.
- Example: For the data {2, 3, 4, 5, 6}, the median is 4. For the data {2, 3, 4, 5, 6, 7}, the median is (4+5)/2 = 4.5.

Important to remember;
The median is not impacted much by outliers and can be used when data can be ranked in order (ordinal data) in a data set.

Measures of central tendency

Mode

- Definition: The value that appears most often in a dataset.
- How to calculate: Identify the value that occurs with the highest frequency.
- Example: In the dataset {2, 3, 4, 4, 5}, the mode is 4 because it appears more than any other value.

Important to remember;
The mode can be used for categorical data and is not impacted by outliers in a data set.

Reflection Exercise

Objectives:

Understand how to use quantitative data to calculate measures of central tendency (Mean, Median, Mode)

Which measure of central tendency reflects the behaviour most accurately?

How does an outlier impact analysis? (Outliers are data points that significantly differ from the data set)

Frequency distribution tables

A frequency distribution table organizes data by showing the frequency, or count, of each value or group of values in a dataset. It consists of columns for the data categories (either individual values or class intervals) and their corresponding frequencies, making large datasets easier to understand and analyze. Optionally, a third column for tally marks can be included to help with counting.

Components of a frequency distribution table

- Category/Variable Column: This column lists the individual values from the dataset.
- Frequency Column: This column shows how many times each category appears in the data.
- Tally Column (Optional): A tally mark column can be used to count the data points for each category as you go through the raw data.

Example 1:

The marks obtained in an English test by a class of 20 students are given below. (Full marks: 30)

25, 23, 28, 21, 24, 23, 23, 25, 22, 21, 28, 21, 23, 25, 28, 21, 25, 24, 22, 22

Represent the above-given marks in a frequency table with tally marks.

Solution:

Marks	Tally Marks	Frequency
21	IIII	4
22	III	3
23	IIII	4
24	II	2
25	IIII	4
28	III	3